Supplementary Material S6
Claim, Citation and Evidence Matrix

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document responds to observation I-08 of the advisory review of 4 November 2026, which asks for every substantive claim of the Discussion and the Conclusions to be traced to its citation and to the evidence that supports it. Each row gives the section of the manuscript, the claim as stated before this audit, the citations or tables it rests on, the evidence located in the repository or in the manuscript, and the status.

The audit covers every sentence of Sections 6 and 7 that states a number, cites a study or uses evaluative, causal or transferability wording, and the one sentence of Section 1 with confirmatory wording. Evidence is located in the extraction matrix (`data/extraction_master.csv`, column named in each row), in the tables of the manuscript and in Supplementary Materials S3 and S4. In the revisions of 12 and 13 November 2026 every row was checked again against the full text of the cited studies (Johnson et al. and Fan et al. on 13 November), whose section or page is recorded in `data/fulltext_coding.csv`; this check also moved Cicceri et al. (2026) to contextual evidence under CE(3), so the counts below refer to 30 eligible studies.

Status: *Supported*, the evidence matches the claim; *Corrected*, the wording or the citations were changed in the manuscript so that the claim matches the evidence, and the evidence column states the change; *Fit check pending*, the claim is consistent with the extracted data but the fit of a citation or a page-level source still has to be checked against the full text.

Totals: 52 claims; 17 supported, 33 corrected and 2 with a fit check pending. In all, 2 rows, 0 of them corrected, still need a citation or page-level check against the full text.

#	Sec.	Claim	Citations or table	Evidence and location	Status
1	6.1	CNN / Deep Learning is the family of 69.6% of the 23 studies with an identifiable ML architecture.	Table 10	`model_family`: CNN / Deep Learning in 15 of 22 (68.2%) after Cicceri et al. left the corpus under CE(3); percentage and denominator corrected.	Corrected
2	6.1	The predominance of CNN reflects the convergence of practical accessibility and methodological maturity.	None	No extracted variable supports an explanation. Reworded as a hypothesis that the corpus cannot test.	Corrected
3	6.1	The highest accuracies (>95%) come from hybrid or ensemble CNN configurations on images without clinical traceability.	Aljbori2025, Hosney2024, Vedantham2025, Alam2025	Table 12: 99.01%, 97.2%, 96.5%, 96.0%. `dataset_category`: Kaggle ASD collections. S4 C.2: 3D-CNN + LSTM, YOLO + CNN, CNN hybrid with GWO, CNN ensemble.	Supported
4	6.1	Hassan et al. report 95.6% (ASD level, DREAM) and Hwang et al. 88.0% (developmental disability screening, own clinical data), both in D4.	Hassan2025, Hwang2024	Full text of Hassan et al.: 95.6% is the training cross-validation score (table II of the article); on the 8 held-out participants the accuracy is 84.0% (table III of the article). Corrected to 84.0%. Hwang et al.: 88.0% confirmed (Results, stratified 5-fold CV).	Corrected
5	6.1	17 of the 31 eligible studies are from 2024 and 2025; none is from 2023.	Table 21	`year`: 2024 (4), 2025 (13), 2023 (0), 2026 (0 after Cicceri et al. left the corpus); denominator corrected to 30.	Corrected
6	6.1	The studies with the highest documented ecological validity integrate temporal modelling or multimodal fusion in real therapeutic contexts.	Hwang2024, Hassan2025, Kiejdo2025	`lim_ecological` = 0 in the three. S4 C.2: LSTM + 2D-FAN, RF + XGBoost on DREAM (three or more sources, Table 16), late fusion of BiLSTM in robot sessions. Hwang et al. is a screening context, not a therapeutic one: reworded to clinical or therapeutic contexts.	Corrected

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7	6.2	The six studies on web repositories without ethical oversight report 99.01%, 97.2%, 96.5%, 96.0%, 88.3% and 84.0%.	Aljbori2025, Hosney2024, Vedantham2025, Alam2025, Kurian2024, Shaik2025	Table 12; `dataset_category`: Kaggle ASD collections (5) and Web videos (1).	Supported
8	6.2	Clinical-origin studies report lower values within each domain: 79.1% (66.9% on full CAFE), 90.7% with 33 children, AUROC 0.86 to 0.92 with 4 children, 88.0% with 89 children and 88.7% with 81 children.	Washington2022, Wang2025, Shi2022, Hwang2024, Chen2024	Table 12 and its notes (Sub. A, f, i); S4 Section A, note 3.	Supported
9	6.2	The 67.8% of Johnson et al. is measured on AffectNet, not on the participants with ASD.	Johnson2025	Table 12, note a; S4 note i.	Supported
10	6.2	Of the five highest accuracies, four come from web repositories and the fifth (Fan et al.) measures specialist-reviewed feedback.	Fan2025, Shaik2025, Hwang2024, Chen2024	Table 12 ranking: 99.01%, 97.6%, 97.2%, 96.5%, 96.0%; S4 note p.	Supported
11	6.2	Hassan et al. reach 95.6% on DREAM, a public dataset curated with ADOS diagnosis and documented parental consent.	Hassan2025	Corrected to 84.0% on 8 held-out participants (table III of the article); 95.6% is the training cross-validation score. Han et al. (95.6%, subject-independent CV) becomes the clinical exception named in the text.	Corrected
12	6.2	Han et al. reach 95.6% with 90 children by fusing EEG and eye-tracking in a laboratory.	Han2022	Table 12; S4 note q.	Supported
13	6.2	Aljbori et al. report 99.01% in the results table and 99.82% in the abstract; Hosney et al. use a dataset created by a co-author; El Rhatassi et al. fall from 94.5% to 53.0%.	Aljbori2025, Hosney2024, Rhatassi2025	S4 notes g and d; Table 12 note; S4 Section B.	Supported
14	6.2	ACC is reported numerically in 20 of 22 studies with supervised metrics (91%), F1 in 11, Recall in 9, AUC in 7 and Precision in 6.	Table 11	Full texts show metrics missing from the matrix: precision and recall of Radočaj and Martinović (table 2 of the article) and Hassan et al. (table III of the article), recall of Hwang et al. (Results) and Silva et al. (sensitivity, table 4 of the article), AUC of Hassan et al. and Silva et al., and F1, precision and recall of Johnson et al. (table 3 of the article). Corrected: ACC 20, Recall 14, F1 12, AUC 9, Precision 9.	Corrected
15	6.2	This distribution can be explained by researchers' familiarity with standard metrics.	None	No study states this reason. Reworded as a hypothesis that the review does not test.	Corrected
16	6.2	AUC improves on accuracy but is insufficient alone under severe class imbalance.	Radocaj2025, Hosney2024	Methodological argument of Section 4.2; both studies recognise class imbalance (Table 8, row 5).	Supported
17	6.2	The most used public dataset is the Kaggle autism collections; the earlier text also named FER-2013.	Table 13	`dataset_category`: Kaggle ASD collections 5 of 31; FER-2013, FERAC, CAFE, DREAM and web videos 1 each. FER-2013 is not among the most used: corrected.	Corrected
18	6.2	Private datasets account for 67.7% of the eligible corpus.	Table 13	`dataset_category`: own clinical 12, clinical eye-tracking 5, robot therapy 2, own non-ASD 1; 20 of 30 (66.7%) after Cicceri et al. left the corpus.	Corrected
19	6.2	No public dataset with Latin American paediatric ASD population was identified in the retrieved evidence.	None	S3 Sections 2 to 12: no dataset documents a Latin American population. The claim is limited to the retrieved evidence.	Supported
20	6.2	Declared limitations: imbalance and sampling 27 of 31, recording and annotation 11 of 31, ecological 10 of 31.	Table 17	`lim_imbalance_sampling` 26, `lim_annotation` 11, `lim_ecological` 9, of 30.	Corrected
21	6.2	Recording inconsistency is due to annotation errors, non-standard protocols and inter-rater variability.	None	Causal wording without extracted evidence. Reworded as the definition of the category declared by the studies.	Corrected
22	6.2	Negative or not determinable MMAT judgements concentrate in representativeness (17 of 31) and confounding (18 of 31).	Figure 2	`mmat_representativeness`: N 9, ND 8; `mmat_confounding`: N 7, ND 10; 17 of 30 in each.	Corrected

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23	6.2	Four studies are negative in both domains: three on web repositories and Fan et al.	Aljbori2025, Hosney2024, Vedantham2025, Fan2025	N in both MMAT columns for exactly these four.	Supported
24	6.2	Seven studies meet the five domains of their category.	He2019, Washington2022, Radocaj2025, Hwang2024, Hassan2025, Leo2018, Wang2025	Y in the five MMAT columns for exactly these seven.	Supported
25	6.2	Values from web-repository studies have no evidence supporting transferability to clinical practice.	Aljbori2025, Hosney2024	MMAT N in representativeness and confounding; Table 5 (no consent, no verified diagnosis).	Supported
26	6.3	Representation: facial or gaze descriptors 16 of 31, end-to-end CNN 12, physiological signals 2, capture format 1.	Tables 14 and 15	`input_group`: 16, 12, 1, 1 of 30; the second physiological study, Cicceri et al., analysed only hand keypoints and left the corpus under CE(3).	Corrected
27	6.3	Two further studies combine face or gaze with electrodermal activity or EEG.	Kiejdo2025, Han2022	Table 16 note (bimodal list); S4 C.2.	Supported
28	6.3	No study reports session-to-session tracking; Hassan et al. (61 children, 3,121 sessions), Shi et al. (4 children, 19 sessions) and four pre-post designs do not meet the definition.	Hassan2025, Shi2022, Silva2021, Lecciso2021, Voss2019, Fan2025, Wall2025	Full text of Silva et al.: seven robot sessions with per-session correct answers and response time for five children (figures 6 to 10 and tables 7 to 10 of the article); S4 coded one session. Voss et al. measure at intake, 6 and 12 weeks (Study Design). Corrected: Silva et al. is the closest study, descriptive and without change quantification; Voss et al. adds a follow-up.	Corrected
29	6.3	No study declares which error it prioritises or reports performance disaggregated by skin tone, sex, severity or comorbidity.	Table 5	Absence recorded during extraction; no study coded with these variables. Page-level verification pending with the full texts.	Fit check pending
30	6.3	Table 22 synthesises the 18 eligible studies with verifiable data provenance.	Table 22	`translational` = 1 in 17 after Cicceri et al. left the corpus.	Corrected
31	6.3	The absence of tools designed for monitoring during interventions is the most significant gap for the Latin American context.	Shaik2025, Hassan2025	S4 C.2: binary ASD/TD and ASD level, examples of the diagnostic orientation.	Supported
32	7.1	None of the 31 studies reports the variables that operationally define therapeutic monitoring.	Section 1, Silva2021	Corrected to: none of the 30 meets the full operational definition; Silva et al. records per-session performance descriptively for five children (see row 28).	Corrected
33	7.1	Primary evidence: 14 studies (D1 10, D2 4); D3 and D4 (17) are comparative context.	Section 2.8	`domain`: D1 10, D2 3, D3 6, D4 11; `primary_evidence` = 1 in 13.	Corrected
34	7.1	Hybrid CNN + LSTM and fusion architectures appear mainly in clinical, robot-therapy and mobile studies; single CNN with transfer learning predominates on public benchmarks.	None	Hybrid part: S4 C.2 lists Hwang, Chen, Kiejdo (clinical, mobile or robot) and Aljbori (web), 3 of 4. Transfer-learning part: the per-study assignment to Table 14 is not yet released (I-03).	Fit check pending
35	7.1	The highest accuracies of the corpus (>95%) come from web-repository studies.	Fan2025	Han et al. (95.6%) and Fan et al. (97.6%) also exceed 95% (the 95.6% of Hassan et al. is a training cross-validation score, row 4). Corrected to four of the five highest, with Fan et al. named.	Corrected
36	7.1	Clinical-origin reference ranges: D1 68.0% to 90.7%, D2 AUROC 0.86 to 0.92, D4 88.0% to 95.6%.	Pliska2025	Table 12: Pliska et al. (D4, clinical) reports 81.3% without validation on unseen data. Corrected: the D4 range is stated for studies with validation and the 81.3% is named.	Corrected

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37	7.1	These models face critical interpretability and phenotypic-heterogeneity challenges.	Hassan2025, Shi2022 replaced by Alam2025, Shi2022, Chen2024	Table 8: interpretability recognised by Alam et al.; generalisation in small samples by Shi and Chen, among others. Hassan et al. recognise neither. The full text of Johnson et al. does not discuss model interpretability (its interpretability refers to how people read expressions, Abstract and Section 1), so Johnson et al. was removed from this sentence and from Table 8, row 2.	Corrected
38	7.1	Only two studies report balanced accuracy and only two report confidence intervals.	Washington2022, Kiejdo2025, Wang2025, Carpenter2020	Table 12 notes e, f and m; Sub. A.	Supported
39	7.1	No evidence allows metric standards for session-to-session monitoring to be set.	Radocaj2025 removed	Full text of Radocaj and Martinović does not address metric standards for monitoring; the citation is replaced by a reference to S4 Section C.	Corrected
40	7.1	DREAM: 61 children aged 3 to 6 with ADOS diagnosis, 3,121 sessions, 306 hours, CC BY-NC-SA 4.0, without facial images.	Hassan2025	S3 Section 7.	Supported
41	7.1	74.2% of the corpus is unimodal and 3 of 31 integrate three or more sources.	Table 16	Full text of Cicceri et al.: one video stream with hand keypoints, not three sources; after its move under CE(3), 76.7% unimodal and 2 of 30 (Hassan et al., Shi et al.) with three or more sources.	Corrected
42	7.1	Fusion of video, physiological and contextual data could improve robustness.	Leo2018, Hassan2025; Han2022 added	Han et al. report 86.67% (eye-tracking) and 81.11% (EEG) against 95.56% fused (S4 note q). Leo et al. integrate AU intensities, head pose and gaze in one facial pipeline (Section 2 of the article): partial fit for the multidimensional nature of facial analysis, not for multimodal fusion.	Corrected
43	7.1	Balancing and partition practice: 2 of 23 declare a balancing technique and 17 omit it; fixed split 12, K-fold 6, no partition 2; 10 of 23 declare no hyperparameter strategy.	Tables 19 and 20	Recomputed on 22: 3 declare a balancing technique and 15 omit it (augmentation over CAFE and FER-2013 to balance expressions in Fan et al., section 3.2.1 of the article); hold-out 13 (Hassan et al. added, 46/8 participants; Johnson et al., AffectNet validation set, sections 3.6.3 and 3.7 of the article), K-fold 6, no partition 2; 9 of 22 declare no hyperparameter strategy.	Corrected
44	7.1	Appropriate balancing can generate substantial improvements in Recall and F1 without changing the architecture.	Chawla2002, He2008 added	No corpus study quantifies it. Reworded as an improvement on the minority class and supported with the methodological sources of SMOTE and ADASYN.	Corrected
45	7.2	Early intervention is the stage in which intensive support has the greatest long-term impact.	None	Uncited clinical claim outside the scope of the review. Reworded as a recognised priority, stating that the review does not assess that evidence.	Corrected
46	7.2	Method repertoire: LIME, SHAP, Grad-CAM, MRMR, SMOTE and ADASYN.	Ribeiro2016, Lundberg2017, Selvaraju2017, Peng2005, Chawla2002, He2008	Primary methodological sources of each method.	Supported
47	7.2	Transcultural validation in Latin American paediatric populations is required.	Shaik2025, Washington2022 replaced by Vedantham2025, Radocaj2025	Vedantham et al. state that the Kaggle images are from children in the United States and Europe (p. 8); Radočaj and Martinović warn that classifiers may underperform on diverse ASD populations (Introduction). Shaik et al. and Washington et al. do not support the statement.	Corrected

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48	7.2	Longitudinal ecological momentary assessment could provide unprecedented information.	Johnson2025, Shi2022	"Unprecedented" removed. Full texts: Johnson et al. name longitudinal studies as future work (Conclusion) and Shi et al. run month-long in-home deployments (section 3 of the article); neither uses ecological momentary assessment, which is now presented as an example and not attributed to them.	Corrected
49	7.2	XAI systems should support, not replace, clinical judgement.	Radocaj2025 replaced by Hwang2024, Hassan2025	Radočaj and Martinović do not address explainability; Hwang et al. apply SHAP (Model interpretation) and Hassan et al. apply Grad-CAM (Saliency Map Approach Findings).	Corrected
50	7.2	Reinforcement learning could enable truly personalised intervention trajectories.	None	"Truly" removed; prospective statement without a corpus claim.	Corrected
51	7.2	Multimodal Transformers can capture subtle patterns undetectable by unimodal analysis.	Washington2022 replaced by Radocaj2025	Reworded as a hypothesis. Washington et al. is unimodal; Radočaj and Martinović propose integrating speech, posture, gaze and physiological signals as future work (Conclusions).	Corrected
52	1	The review by Baños et al. confirms that the face is the predominant modality.	Banos2024	"Confirms" replaced by "reports": the statement summarises another review and is not verified by this one.	Corrected

Notes.

Section numbers follow the manuscript: 1, Introduction; 6.1 to 6.3, Discussion; 7.1 and 7.2, Conclusions and future research.

Citation keys are those of the bibliography of the manuscript. Every citation added or replaced in this audit is named in the citations column of its row.

On 13 November 2026 every citation group of the manuscript was sorted by reference number. In the running text and notes, groups that support a statement have at most two references; groups that enumerate every study of a stated set (for example, the studies of a domain) keep the complete list.

Rows with a fit check pending require the full text of the cited study; they are resolved together with observations I-01 and I-11.